

Android Devices in the Market

- Smart phone
- Tablets
- E-reader devices
- Internet TVs
- Automobiles
- Smart watches

* Android Environmental Setup

Require Tool

- Start android application development on following OS

- Microsoft windows xp
- Mac
- Linux

- Required Tool to develop Android

- Java JDK
- Android SDK

1. Step 1: Setup Java Development kit (JDK)

→ download latest version of JDK

2. Step 2: Android Studio Setup

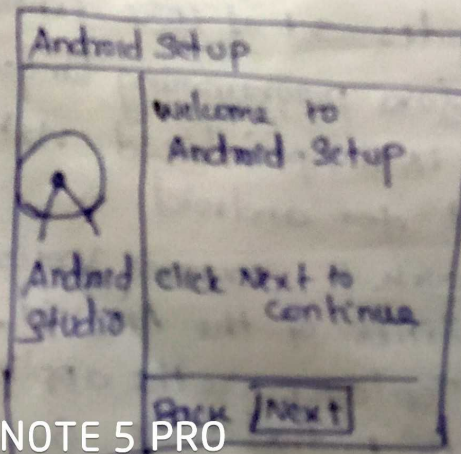
Google Android Studio

<https://developer.android.com/studio>

Step 3:

(Download) → click download

3. click next to Android Studio Setup



Step 4: Pick the components of Android Studio which want to install from the screen, click Next.

The diagram shows a window titled "Android Studio Setup" with a tab labeled "choose components" (indicated by a circled 'A'). Below the tab is the text "Select components to install". To the right of this text is a list of three items, each with a checkbox: "Android Studio" (unchecked), "Android SDK" (checked), and "Android virtual device" (checked). At the bottom of the window are three buttons: "Back", "Next", and "Cancel".

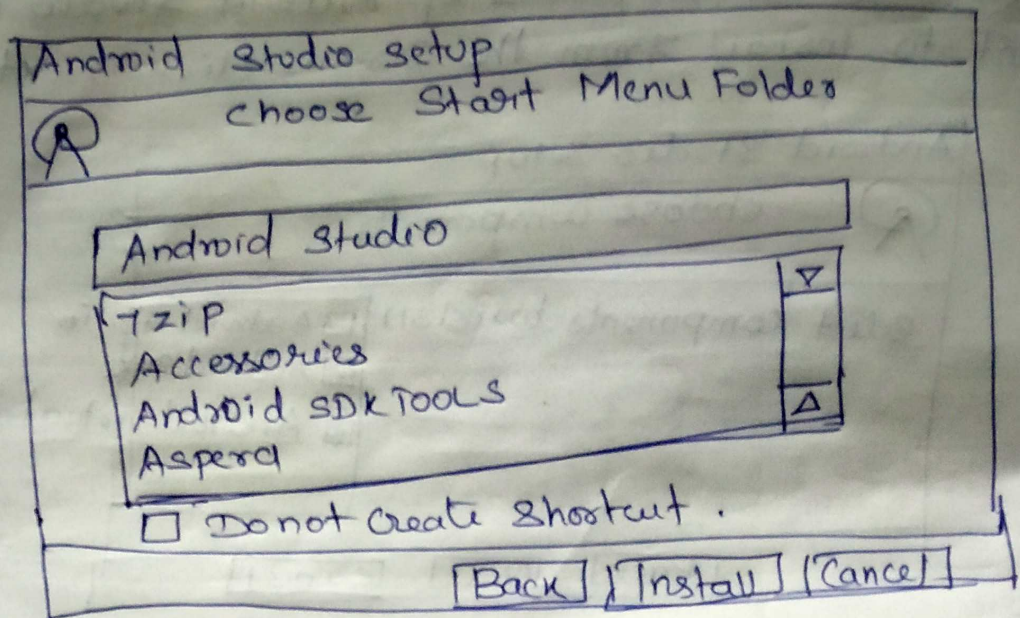
Step 5: click next in Configuration Setting

The diagram shows a window titled "Android Studio Setup" with a tab labeled "Configuration Settings" (indicated by a circled 'A'). Below the tab is the text "c:\Program Files\Android\Android Studio". At the bottom of the window are two buttons: "Back" and "Next".

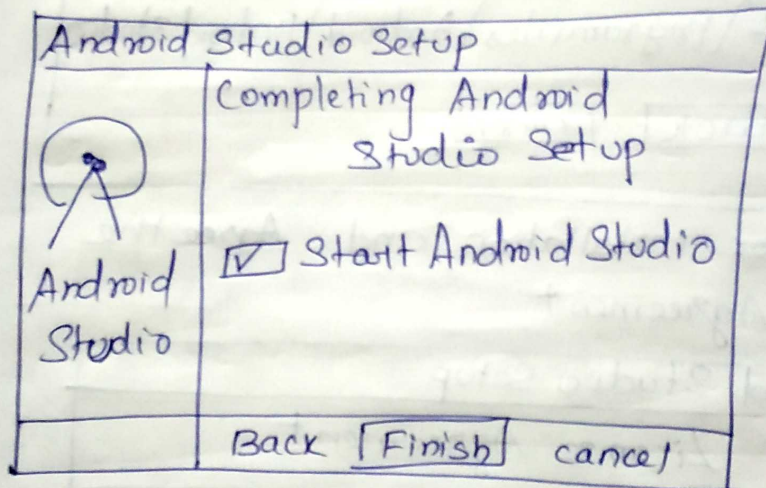
Step 6: wait for installation and Agree the License Agreement

The diagram shows a window titled "Android Studio Setup" with a tab labeled "License Agreement" (indicated by a circled 'A'). Below the tab is the text "If you accept terms and conditions click I agree.". At the bottom of the window are three buttons: "Back", "I Agree", and "Cancel".

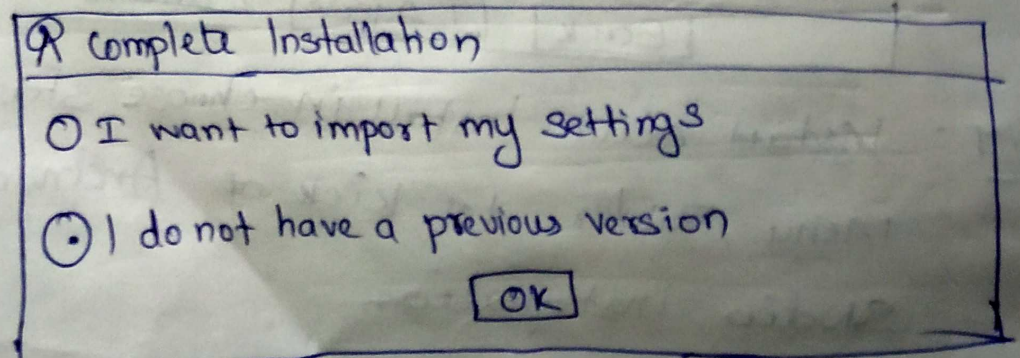
Step 7: ~~Install~~ click Install in choose start menu Folder to kick off Android Studio installation



Step 8: Installing Android Studio, when install completed Completing Android Setup Screen will open, click Finish button



Step 9: Here either import settings from a previous version of Android Studio or continue with new setting, click ok



Step 10

Android Studio Setup wizard

☒ Standard
☐ Custom

Previous **Next** Cancel

Click Next to continue.

Step 11: Android SDK need to download

- It contains all of the packages and tools require to develop a functional Android

Verify Settings

Current Setting

Android Emulator 363M13
Android... SDK Build tools 33 53M13
...

Prev **Next** Cancel

Click next

Step 12:

~~SDK~~ Quickfix Installation

☒ License Agreement

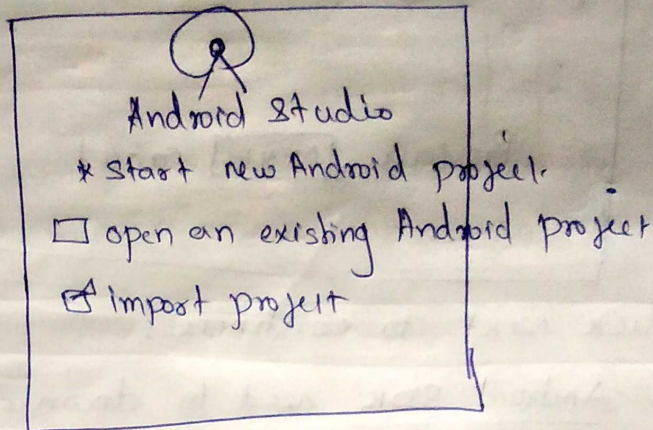
Licenses
- android-sdk
- Android SDK Platform

☐ Decline ☒ Accept

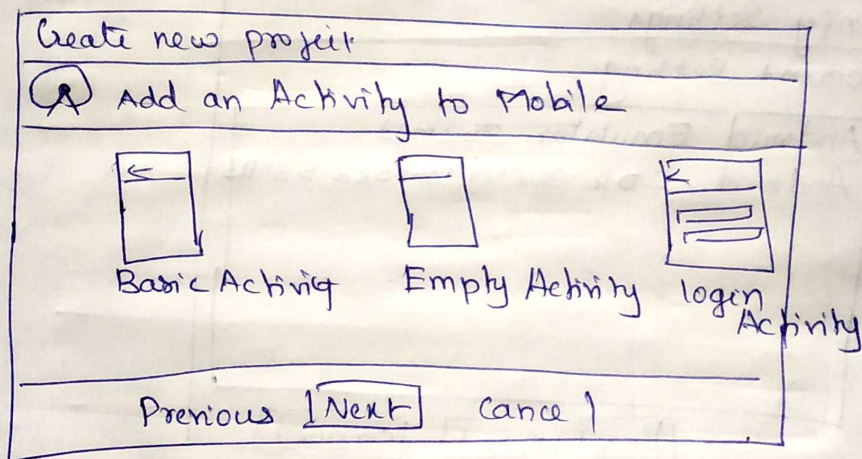
Next

Click Next.

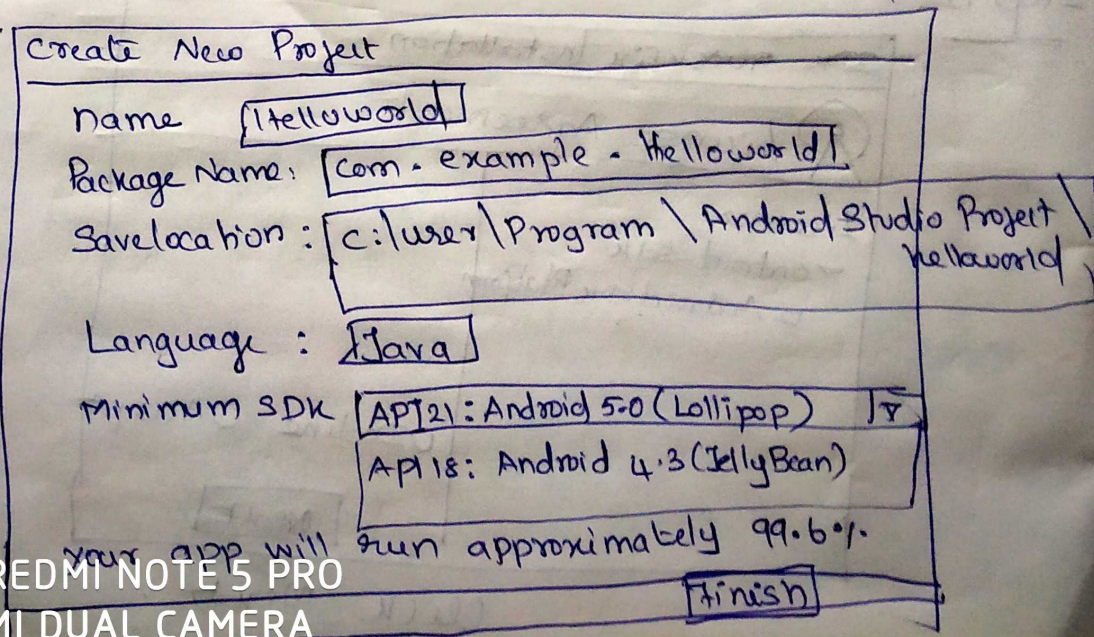
Step 13. For Creating Android Virtual Devices (AVDs)
Start new project wizard in Android Studio project



Step 14: Set up Hellow world project, Add an activity to Mobile.



Step 15 Create new project named Helloworld



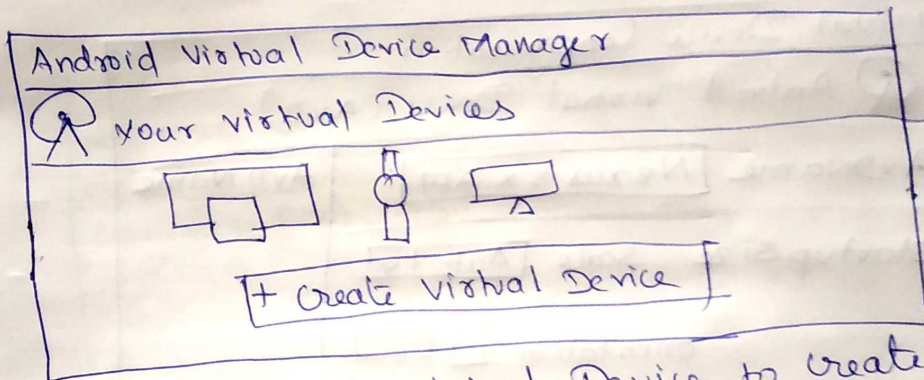

```

Helloworld
File Edit view Navigate Analyse window
Helloworld > app > src > main > java > Main Activity.java

Android >
package com.example.helloworld;
import .....
public class MainActivity extends
    AppCompatActivity
@override .

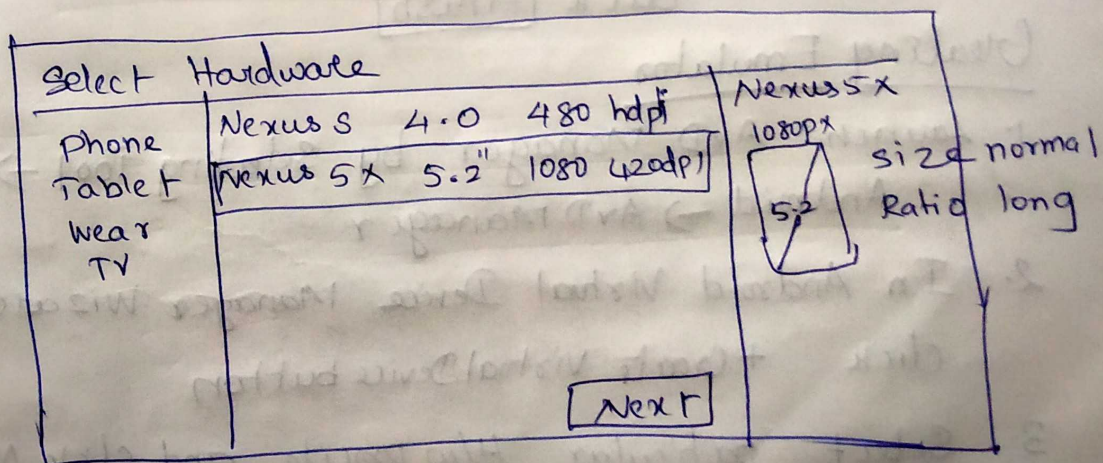
```

Launch AVD Manager by selecting
 Tools $\Rightarrow$ Android $\Rightarrow$ AVDManager or
 AVD Manager button from Toolbar.



click + Create virtual Device to create
 new AVD

Select hardware profile and click Next



click Next

Virtual Device Configuration System Image

Recommended ☒ x86 Image ☐ other image

Name	level	ABI	Target	API Level
Lollipop	23	x86_64	Android 6.0	N
Marsh				
Jelly Bean	18	x86	Android 4.3	Android
Jelly Bean	17	x86	Android 4.2	System Image
				x86_64

Prev ☒ Next ☐ Cancel

click ☒ Next

Virtual Device Configuration

☒ Android virtual Device (AVD)

AVD Name ☒ Nexus 5x API

Startup Size scale ☒ Auto ☐ x

orientation ☐ ☐

Portrait Landscape

☒ Finish

click ☒ Finish

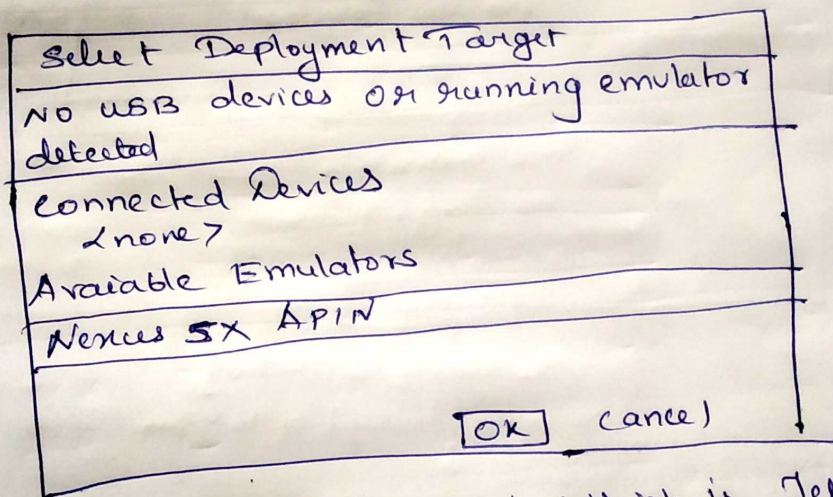
Creating Emulator

1. Launch AVD Manager by selecting Tool → Android → AVD Manager
2. In Android Virtual Device Manager Wizard click + Create Virtual Device button
3. Select particular H/w profile and click Next
4. Select Jelly Bean from list of image and click download.

5. Accept the agreement, download SDK
6. After SDK has download, click jelly Bean again and click next.
7. click finish Button.

* Launching Your First Android Application

- // Here First create Hellow world! app.
- // Launching your first Android Application
1. Select Run → Run app from Android studio Menu bar



2. Select Nexus 5X API 21 which is Jelly Bean Emulator
3. Emulator fully loaded and started
Android studio install Hellow world Application

